NEPTA MOCK EXAMINATIONS – 2026

UACE PHYSICS

PAPER 1

PROPOSED SCORING GUIDE

Item 1

(a) ***The maximum safe speed along the circular bend for no skidding***

$V_{max} = \sqrt{rg\ tan\theta} = \sqrt{50x9.81\ \tan 30} = 16.8283\ ms^{-1}$

The maximum safe speed on the circular bend is $16.8283\ ms^{-1}$

The outlet pipe was fixed at 10.0cm from the bottom so as water to move out at maximum pressure and speed since pressure increases with depth and the depth is proportional to the velocity at which water emerges out of the pipe..

Velocity of emergence, $V = \sqrt{2gh}$

$V = \sqrt{2x9.81x3.6} = 8.4043\ ms^{-1}$

Hence water will be ejected through the pipe at $8.4043\ ms^{-1}$

During the off loading:

Work done using a wooden plank $= Fxd = mgsin\theta\ x\ length\ of\ the\ plank$

$W.D = 250x9.81 \sin\left(\frac{1.5}{7}\right) \times 7 = 64.206J$

Work done vertically $= mgh = 250x\ 9.81x\ 1.5 = 3{,}678.75J$

More work is done in lowering the tank from the truck than offloading it using a wooden plank so the son's proposal is better since less work is done.

The noise during offloading the tank was due to friction between the surface of the tank and the wooden plank

Origin of friction

The actual area of contact between the wooden plank and the surface of the tank is very small, this creates high pressure at contact points. This pressure pushes the contact points to form cold

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welds. The cold welds have to be broken for motion to occur which creates friction between the tank and the plank.

How to minimize friction:

- Through lubrication using oil, grease between the two surfaces
- Using rollers instead of sliding
- By streamlining shapes/surfaces

(b) During flight:

The airplane wings are curved on top and flatter on bottom. This allows air move faster over the top and with low pressure, air moves slower below with high pressure. The pressure difference creates a lift force that pushes the plane up.

Chimneys: Wind blowing fast over a chimney top creates low pressure which helps to pull smoke up out of the chimney to the surrounding areas.

Spinning ball: Air above the ball moves with fast moves with fast speed which creates a low pressure than the air below if which moves at a low speed with high pressure below by Bernoulli's principle. The pressure difference causes the spinning force on the ball.

Item 2

***(a) Analyzing Paul's Concern** About Image Size exceeding Screen Size*

Area of the square screen = 1600 cm^2

Length of sides of the screen = $\sqrt{1600}$ = 40.0 cm

Maximum allowable deflection from the center = 40.0÷2 = 20.0 cm = 0.2m

Distance from plates to screen, L = 20.0 cm = 0.2 m

Length of deflecting plates, l = 5.0 cm = 0.05 m

Distance between plates, d = 2.0 cm = 0.02 m

Accelerating (tube) voltage, V_a= 240 V

Maximum deflecting voltage, V_d = 165 V

The total deflection Y on a screen is given by:

$Y = \frac{eV_d\, lD}{md\left(\frac{2eV_a}{m}\right)} = \frac{V_d lD}{2dV_a}$ (*where D is the distance from the center of the plates to the screen*)

$$Y = \frac{165 \times 0.05 \times 0.225}{2 \times 0.02 \times 240} = 19.34cm$$

Conclusion: Since the deflection is 19.34 cm and ***less than*** the maximum screen limit from the center of 20.0 cm then the image fits completely within the screen boundaries thus *Paul's worry about the images being too big is not justified.*

Concern About Dangerous Radiations

When accelerated electrons strike the screen or internal metal components, their kinetic energy is converted into **x-rays** and **heat.**

From $hf_{max} = eV_a \Rightarrow f_{max} = \frac{eV_a}{h} = \frac{1.6\times10^{-19}\times240}{6.63\times10^{-34}} = 5.79 \times 10^{16} Hz$

Conclusion: Since the frequency of the radiations is 5.79×10^{16} Hz and greater than the threshold of 3×10^{15} **Hz** thus life-threatening X-ray radiations are produced during operation.

Safeguards for Users

***Lead-Glass Shielding*:** The front screen must be built using thick glass infused with lead to absorb the emitted X-rays before they reach viewers.

***Internal Casing*:** Line the inside of the TV box with thin metallic shielding (like steel or lead sheets) to contain stray radiations.

(b) Helping John Understand Solar Power for the TV Set

The sun's radiation can power the TV through **solar photovoltaic technology**.

Photovoltaic Effect: Solar panels are made of semiconductor materials (such as silicon). When sunlight hits the panels, photons transfer energy to electrons in the material.

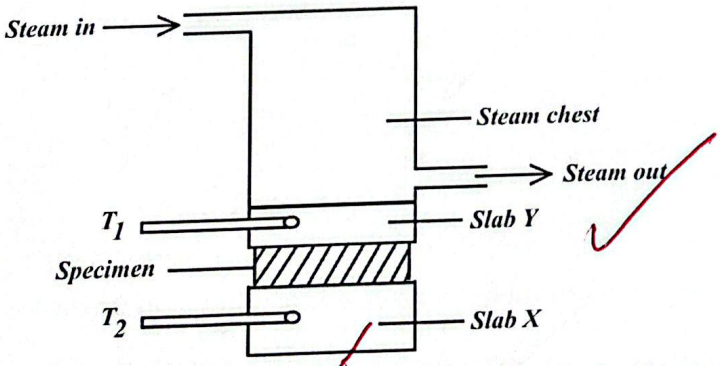
Generation of Electricity: This energy knocks electrons free, creating free electrons and holes. An internal junction structure guides these electrons into a directed stream, creating Direct Current (DC) electricity.

Storage and Regulation: Because solar intensity changes throughout the day, the generated electricity goes through a charge controller to safely charge a rechargeable battery. The battery ensures stable power supply even at night.

Powering the Prototype: Since the TV project is meant to be highly affordable, it can be designed to run directly on the low-voltage DC power from the battery, completely bypassing the need for a costly inverter or connection to the national grid.

Item 3

(a) Determining Thermal Conductivity of a Poor Conductor (Clay)



- The material is cut into a thin disc of known cross-sectional area and thickness and then placed between the steam chest and brass slab, X.
- Steam is passed through the steam chest until steady states are attained and the steady temperatures θ_1 and θ_2 are read and recorded from the thermometers T_1 and T_2.
- The specimen is removed and slab X is heated directly to a temperature 10°C above θ_2.

- Steam chest is removed and specimen disc is directly placed onto the slab X. The temperature fall is recorded in equal time intervals.
- A graph of temperature against time is plotted and a tangent is drawn at θ_2 and slope, S is determined.
- The rate of thermo conductivity, K is then calculated from $\frac{KA(\theta_1-\theta_2)}{x} = mcS$

where c is s.h.c of brass and m is mass

Solar Panel & Solar Constant Calculations

$$Solar\ constant = \frac{P_s}{A} = \frac{\delta A_s T^4}{4\pi R^2} = \frac{\delta(4\pi r^2)T^4}{4\pi R^2} = \frac{\delta r^2 T^4}{R^2}$$

$$Solar\ constant = \frac{5.67 \times 10^{-8} \times (7.0 \times 10^8)^2 \times (5700)^4}{(1.5 \times 10^{11})^2} = 1303.455 Wm^{-2}$$

$$Total\ power\ received\ by\ the\ solar\ concentrator = solar\ constant \times Area = 1303.455 \times A$$

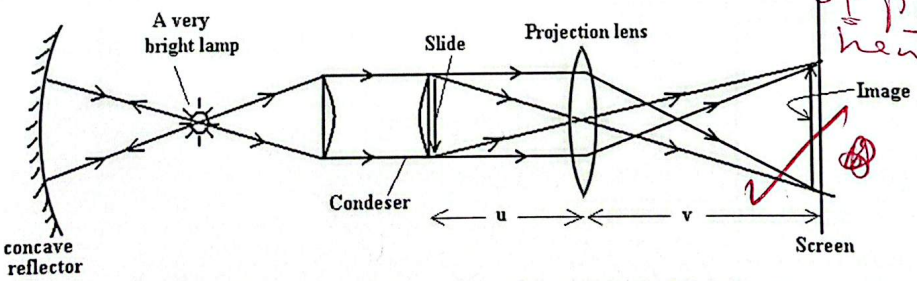
$$300 = \frac{22}{100} \times 1303.455A$$

$$A = 1.046m^2$$

The low and high sound is due to interference

- The low sound is due to destructive interference and high sound is due to constructive interference.
- When the sound waves from the loudspeakers meet in phase, when the path difference is equal to an integral multiple of the wavelength, constructive interference occurs hence a loud sound.
- When the path difference is not equal to an integral multiple of wavelength, destructive interference occurs hence a low sound.

(b) The suggested device is a projector

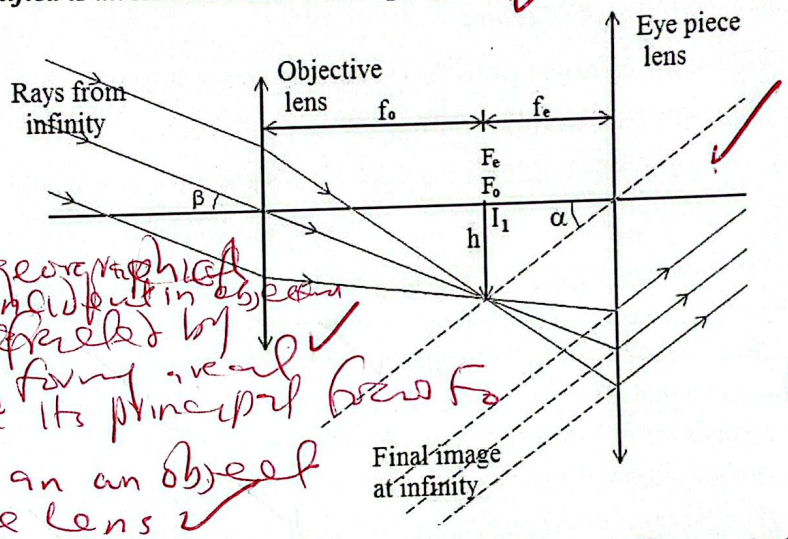


- A lamp is placed at the center of curvature of the mirror which illuminates the condenser lens.
- The concave mirror concentrates and focuses the light into the condenser which then illuminates the object slide uniformly. The projection lens then inverts the image and focuses it onto the screen.
- A real magnified upright image is then formed on the screen.

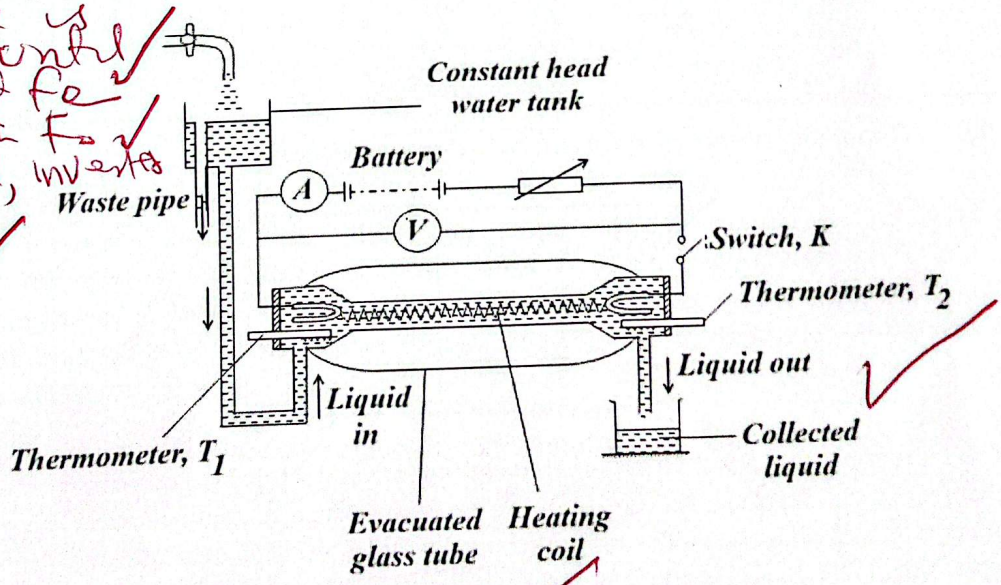
(a) Greenhouse Effect

- Radiations from the Sun have long and short wavelengths.
- Short wavelength radiations penetrate the ozone layer and long wavelength radiation are absorbed.
- The radiations that reach the bus penetrate through the roof of the bus and are absorbed by the chairs and people inside it.
- They are again re-radiated as long wavelength radiations which are then trapped inside, hence raising temperature in the bus.

The device identified is an Astronomical Telescope operating in normal adjustment



Determining Specific Heat Capacity of the liquid using Continuous Flow Method



- The liquid is made to flow at a constant rate.

- Switch K is closed and the liquid is heated until steady state is attained.
- The inflow and outflow temperatures θ_1 and θ_2 are read and recorded from the thermometers T_1 and T_2.
- The ammeter reading, I_1 and voltmeter reading, V_1 are read and recorded.
- The mass m_1 of the liquid collected in time t is measured and recorded.
- The flow rate and rheostat are adjusted and liquid is heated until the inflow and outflow temperatures are the same.
- The new ammeter and voltmeter readings I_1 and V_2 are read and recorded.
- The mass m_2 of the liquid collected in the same time t is measured and recorded.
- The s.h.c of the liquid is then calculated from $c = \frac{(I_1V_1 - I_2V_2)t}{(m_1 - m_2)(\theta_2 - \theta_1)}$

Specific Heat Capacity of the Coolant

Heat lost by the solid = Heat gained by the calorimeter and coolant(liquid)

$C\Delta\theta = m_L c_L(42 - 20) + m_c c_c(42 - 20)$

$276(80 - 42) = 0.1 \times c_L(42 - 20) + 0.2 \times 400(42 - 20)$

$c_L = 3967.27 Jkg^{-1}K^{-1}$

Conclusion / Evaluation: Since 3967.27 J kg^{-1} K^{-1} is greater than 3740 J kg^{-1} K^{-1}, the liquid is a good coolant.

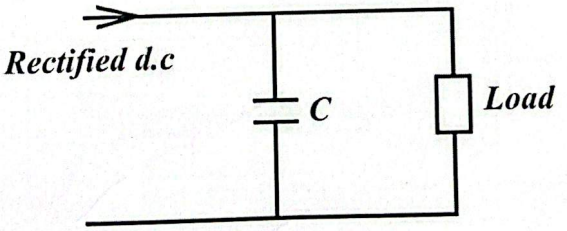
(b) The device used is a radar speed gun

- Electromagnetic radiations from the gun are directed onto the car.
- These radiations are reflected and received by the gun.
- The change in wavelength and frequency is detected by the gun.

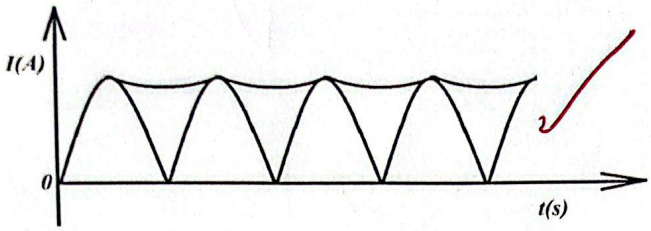
The speed of the car is determined from $v = \frac{c\Delta f}{2f}$, where f is the frequency of electromagnetic radiation

Item 5

(a) Smoothing the rectified d.c current using a capacitor



The capacitor is connected across the load and smoothens rectified d.c current by continuous charging and discharging. As voltage increases, the capacitor charges and some energy is stored in the electric field associated with it. As the p.d reduces, the capacitor discharges and energy is transferred to the load. This maintains the p.d and current across the load and the output p.d is as shown below.



Efficiency of the motor

$$Efficiency = \frac{E_b}{V} \times 100\%$$

$$E_b = V - Ir = 90 - 1.85 \times 5.2 = 80.38V$$

$$Efficiency = \frac{80.38}{90} \times 100\%$$

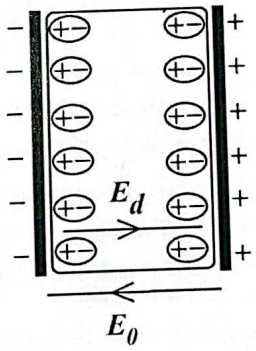
$$Eff = 89.3\%$$

They should install another motor since the efficiency is less than 90%

Why efficiency is always less than 100%

- Some power is lost as heat due to resistance in the armature coil
- Some energy is lost due to eddy currents as magnetic flux changes
- Hysteresis loss due to internal friction as magnetic flux changes

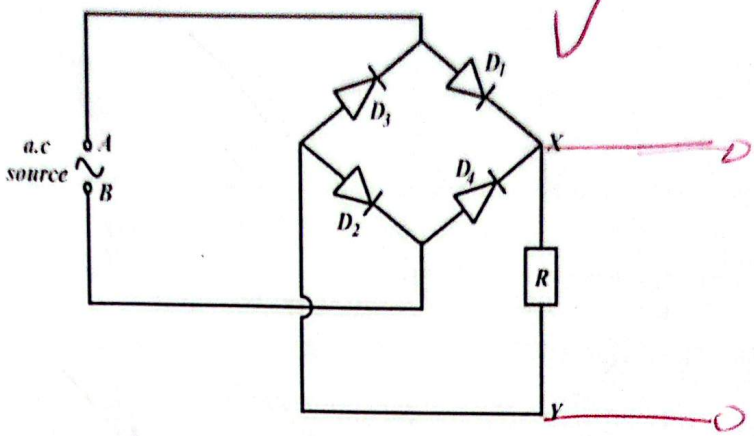
How the insulator works



When the insulator/dielectric is placed between the plates of the capacitor, its molecules get polarized creating a positive charge near the negative plate and a negative charge near the positive plate. These are bound charges and cannot be conducted thus creating an electric field intensity E_d in opposite direction to that of the applied electric field intensity E_0 which reduces the effective electric field intensity $E = E_0 - E_d$.

From $E = \frac{V}{d}$, reduction in E shows a reduction in V and from $C = \frac{Q}{V}$, a reduction V leads to an increase in C hence the capacitance increased when a dielectric is inserted between the plates of the capacitor.

Role of diodes.



The bridge of diodes acts as a full wave rectifier to covert alternating current to direct current.

In the half cycle when A is positive, diodes D_1 and D_2 are forward biased while D_3 and D_4 are reverse biased. The current from A through D_1, X, Y, D_2 and back to B.

In the next half cycle when B is positive, diodes D_3 and D_4 are forward biased while D_1 and D_2 are reverse biased, current flows from B through D4, X, Y, D_3 and back to A.

In both half cycles, current flows through the load in the same direction XY hence the D.C out put

(b) How the ignition system works.

The ignition system relies on self-induction to generate high voltage required to produce a spark at the spark plugs.

When the ignition switch is turned on, current flows through the primary winding of the ignition coil which creates a magnetic field around the coil. When the current is suddenly interrupted, the collapsing magnetic field induces a high voltage in the secondary winding of the ignition coil.

The high voltage is then sent to the spark plugs creating a necessary spark to ignite the air fuel mixture in the engine cylinder. Increase in the temperature expands the air fuel mixture and exerts pressure to the piston which runs the crankshaft that runs the engine.

Item 6

(a) Working of the metal detector

It consists of an inductor, a capacitor and a resistor connected in series with an a.c source and adjusted so that the capacitive reactance is equal to the inductive reactance of the circuit and thus set at resonance such that current through the circuit is maximum. When a metal is inserted into (passes through) it, the changing magnetic field in the circuit coil(inductor) links with the metal creating eddy currents. The eddy currents oppose the original magnetic field and thus affect the inductive reactance of the circuit. This in turn results into a change in current flowing through the circuit which triggers the alarm connected to the resonance circuit.

Specifications of the inductor/coil

$X_L = X_C$ ✓

At resonance, $f_0 = \frac{1}{2\pi\sqrt{LC}} \Rightarrow L = \frac{1}{4\pi^2 f_0^2 C} = \frac{1}{4\pi^2(90.9\times10^6)^2\times5.47\times10^{-12}}$ ✓

$L = 5.604 \times 10^{-7} H$ ✓

Conclusion/Advice: The machine requires a coil of self inductance $5.604 \times 10^{-7} H$ in order to be fixed to work again.

(b) time taken by the portable to run down

In parallel arrangement, $C = C_1 + C_2 = 0.36 + 0.36 = 0.72kF = 0.72 \times 1000 = 7200F$ ✓

Energy stored by the capacitor bank, $E = \frac{1}{2}CV^2 = \frac{1}{2} \times 7200 \times 100^2 = 3.6 \times 10^6 J$ ✓

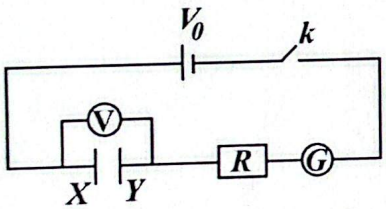
But $Energy, E = Power \times time = Pt$

$t = \frac{E}{P} = \frac{3.6 \times 10^6}{1000} = 3600s$ ✓

In hours, $t = \frac{3600}{3600} = 1hour$ ✓

Conclusion /Advice: The loudspeaker can only work for 1 hour (less than 2 hours needed) when fully charged and therefore does not meet James' need. ✓

Charging of the capacitor bank



The capacitor plates XY are connected in the circuit as shown above. When the switch, k is closed, electrons flow from the negative terminal of the charging battery to plate Y of the capacitor bank connected to it and the galvanometer deflects to the maximum. At the same time, electrons from the other plate X of the capacitor flow towards the positive terminal of the cell making it positively charged. The negative charges which accumulate on plate Y oppose further flow of electrons from the battery hence their speed reduces and the charging current gradually reduces to zero and the galvanometer deflection goes to zero. At this point, the p.d across the plates of the capacitor bank is equal to that of the charging battery, V_0 and so it is fully charged.